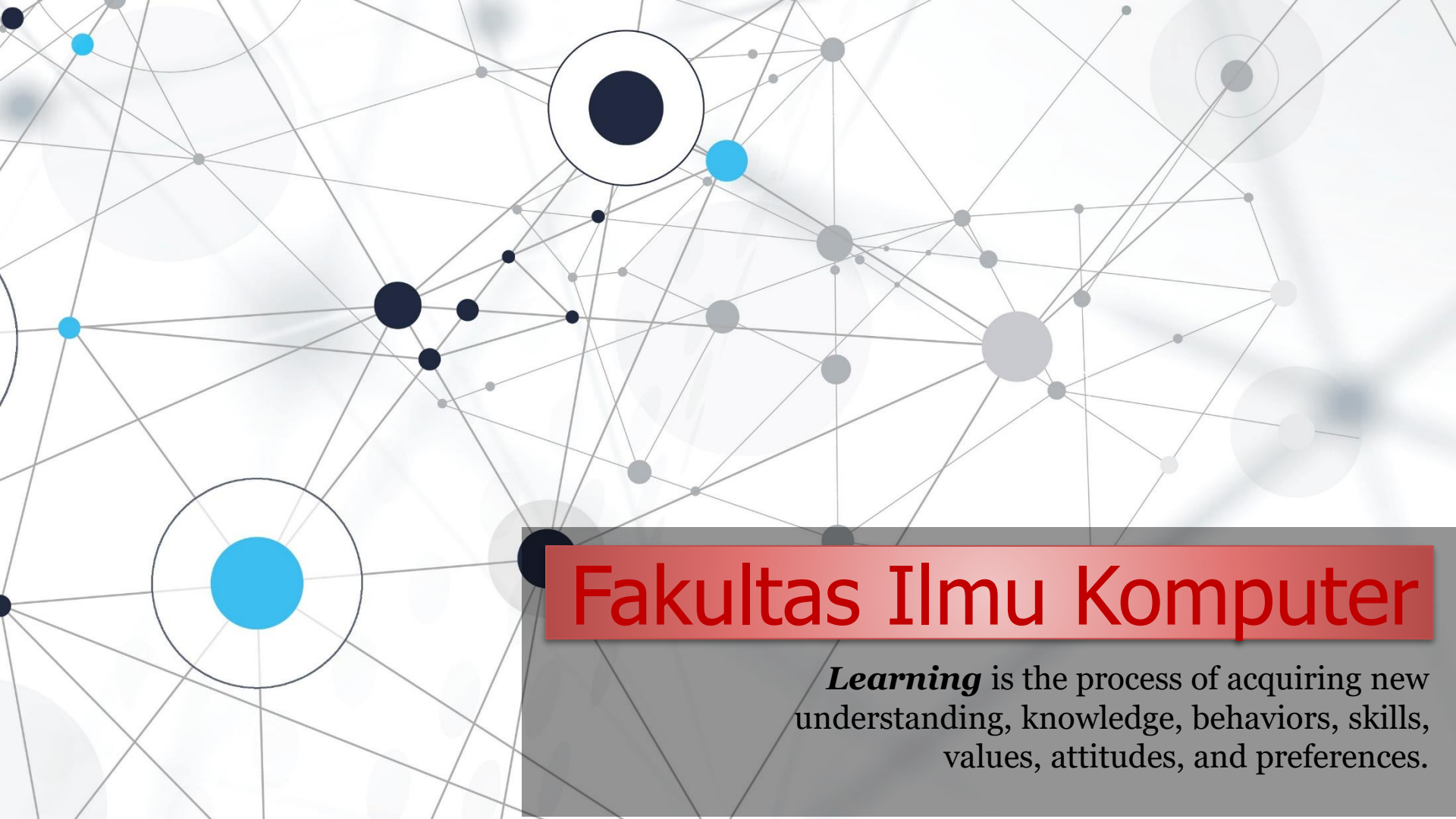
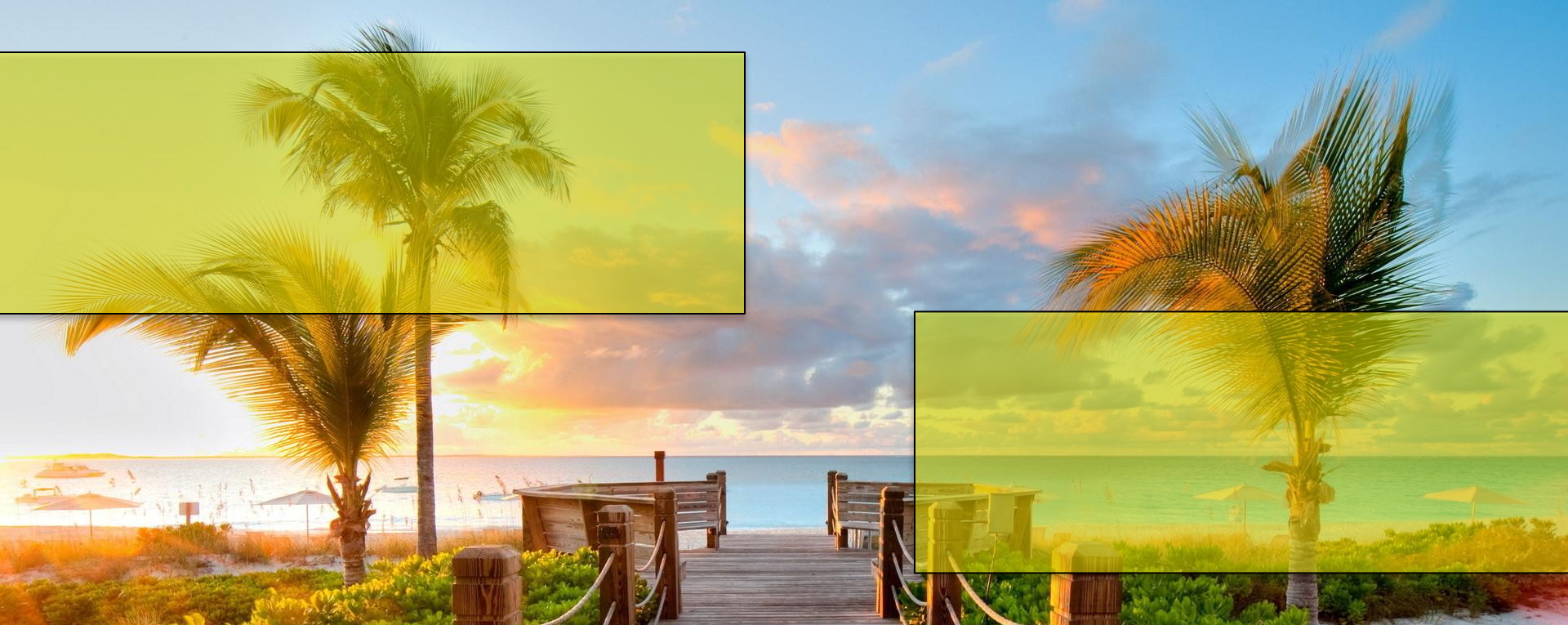


An abstract network diagram with various sized nodes (black, blue, and grey) connected by thin grey lines. Some nodes are highlighted with larger circles. The background is white with faint grey circles.

# Fakultas Ilmu Komputer

***Learning*** is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

## Course Information & Overview



# Perkenalan dari mahasiswa

Sebutkan:

- (1) Nama lengkap & Jurusan
- (2) Daerah asal
- (3) Cita-cita (Contoh: programmer, web designer, *etc.*)
- (4) Advanced teknologi (tecknologi terbaru) apa yang kalian suka?
- (5) Ekspektasi (*harapan*) anda, pada saat/sesudah mengambil class ini?
- (6) Sudah/belum pernah ambil class saya?



# **Welcome *to* the Back-end Web Development *course***



# Instructor Info & Contact

**Full Name** : Semmy Wellem Taju  
**E-mail** : [semmy@unklab.ac.id](mailto:semmy@unklab.ac.id)  
**Social Media** : <https://web.facebook.com/sadewastw>



## Instructor information

- **Office** : Universitas Klabat
- **Mobile phone** : 0811-8432-447
- **Office Hour** : individual appointment

# Prerequisites (*Prasyarat*)

The student must have some *prerequisite skills* and required as a *prior condition*.



- ☐ Data structures
- ☐ A good working knowledge of PHP and Python programming
- ☐ SQL (*database*)
- ☐ Web server and development
- ☐ No prior *knowledge* of design is assumed.

# Student Conduct Code

*(framework for student behavior)*



- ✓ Students will practice high standards of professional *honesty* or *integrity* (*quality of being honest and having strong moral principles*).

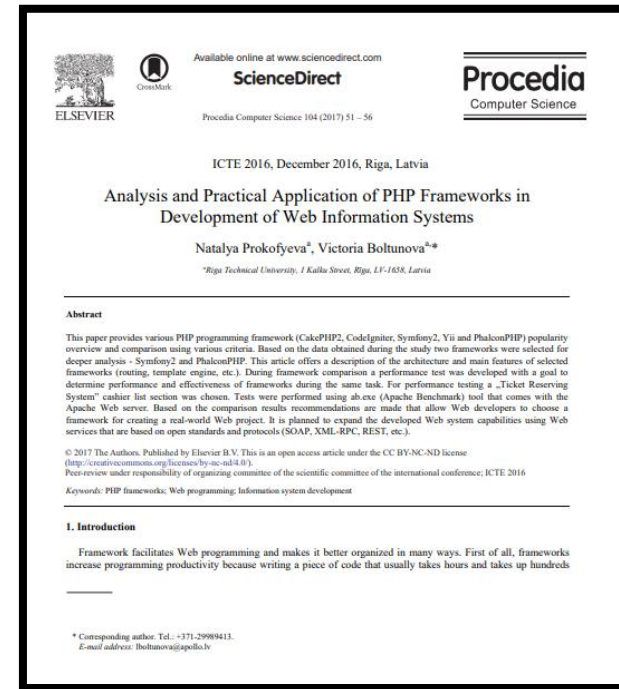
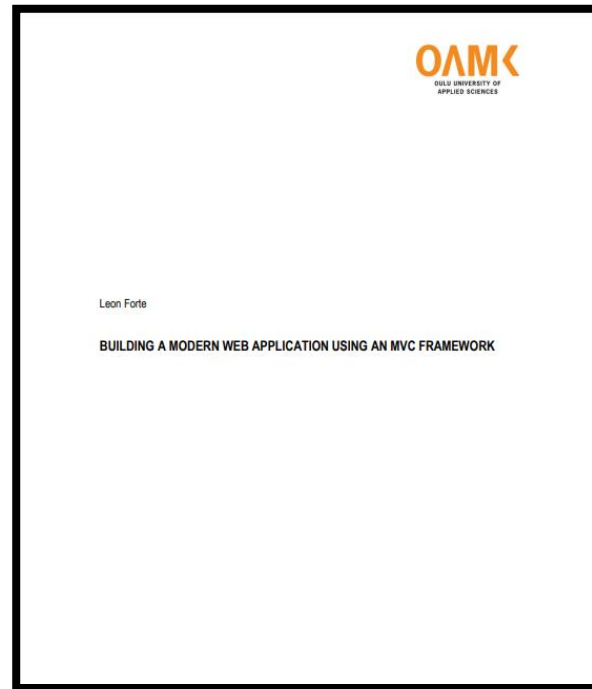
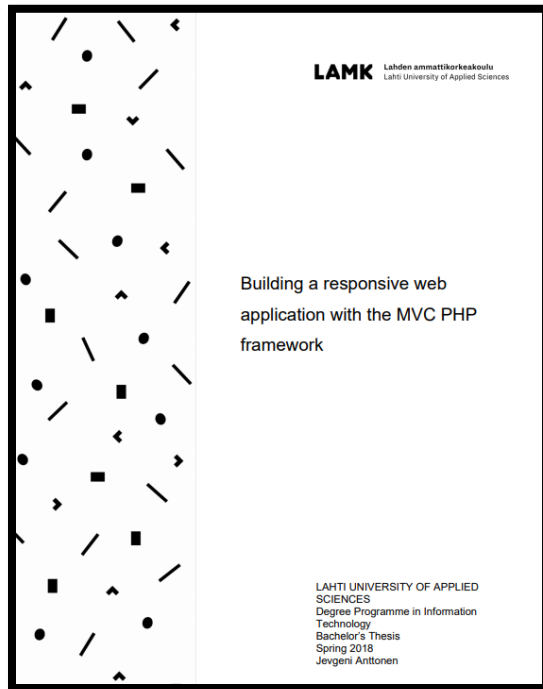


- ✓ In particular, all work (*homework* and *exams*) is to be ***your own individual work***.



- ✓ You may discuss programs but NOT *copy* any code, file or document from any source (*friend* and *online repository*).

# Course Textbook & Recommended Readings



Anttonen, J. (2018). *Building a responsive web application with the MVC PHP framework.*

Forte, L. (2016). *Building a Modern Web Application Using an MVC Framework.*

Prokofyeva, N., & Boltunova, V. (2017). *Analysis and practical application of PHP frameworks in development of web information systems.* Procedia Computer Science, 104, 51-56.

# Grading & Evaluation

We will have *mid* and *final exams*, so Grading is based on *mid* score, Quiz, class activity, *homework/assignments*, *attendance* and *final project* only.

## ❑ Grading



- **Mid-term exam:** 20%
- **Homework/assignments:** 15%
- **Class activity (1-7):** 15%
  - Programming assignments / words
- **Quiz:** 10%
- **Attendance:** 10%
- **Final Project:** 30%
- **Bonus:** Class participation and Student Conduct (*professional behavior & honesty*): 5%



## ❑ Important Notes

- You may **fail** this course if you miss (*more than*) 7 classes of the whole classes.
- Academic dishonesty (e.g. *cheating, plagiarism, and etc.*) will be taken seriously and heavy penalty can be imposed.





---

Apa yang *di* maksud dengan  
**Web Development?**

Apakah menjadi seorang  
**Web Developer** saja  
sudah cukup di era  
digital skarang ini?

# Back-end Web Development Course



Back-end Web development (*pengembangan web*) adalah sebuah proses pengembangan (*develop*) dan pemeliharaan (*maintain*) situs web.

Pastinya, jenis pekerjaan ini (*back-end developer*) dilakukan di balik layar (*behind the scenes*) dengan tujuan untuk membuat situs web yang tampak menarik (*attractive website*), dan dapat bekerja dengan cepat.

# Jenis-Jenis Web Development

## FRONT-END WEB DEVELOPMENT

### Tugas utama:

- ❖ Seorang front-end web developer bertugas untuk membangun UI yang dapat membantu pengguna (*users*) untuk mencapai tujuan di dalam pengembangan situs.
- ❖ **Front-end web developer** juga sering ikut dalam aspek perancangan User Experience (UX) dalam project.
- ❖ Memiliki *background* di bidang UX dapat menjadi nilai plus bagi seorang front-end developer.



## BACK-END WEB DEVELOPMENT

### Tugas utama:

- ❖ Seorang back-end developer bertugas untuk mengelola server situs web, program (*database*), dan software, dengan tujuan agar semua fitur tersebut dapat bekerja dengan baik.
- ❖ **Back-end developer** bekerja dalam sistem seperti **server**, *sistem operasi*, **API**, dan *basis data*.
- ❖ Bertugas untuk mengelola *code* dalam situs untuk keperluan *keamanan*, *konten*, dan *struktur situs*.



# Full Stack Developer

**Full Stack Developer** adalah sebuah profesi di mana seseorang memiliki keahlian untuk memperbaiki front-end (***client side***) maupun back-end (***server side***) dari sebuah aplikasi atau website. Karena berhubungan dengan dunia IT, ada beberapa *skills* yang wajib dimiliki oleh seorang Full Stack Developer.

Kemampuan menguasai *front-end*, *back-end*, UI dan UX, serta me-manage database adalah *hard skills* yang wajib dikuasai oleh seorang Full Stack Developer.



# Perbedaan Umum Front-end & Back-end ?

Berkaitan dengan proses web development, **front-end** adalah apa yang pengguna lihat pada tampilan depan sebuah website dan **back-end** adalah sistem di balik layar yang mengolah *database* dan juga *server*.

**Kenali istilah berikut ini:** (1) bagian front-end sering pula disebut sebagai "*client-side*" dan (2) back-end disebut dengan "*server-side*".

**Kenyataannya, *developer* hanya mengikuti system/service yang sudah ada dari sebuah company.**

**Jadi, dibutuhkan keahlian yang sesuai dengan kebutuhan dari company tersebut.**

# Perbedaan pada skill yang harus dikuasai

**Seorang Front End Developer** harus memiliki kemampuan dasar yaitu minimal menguasai tiga bahasa pemrograman seperti Javascript, HTML dan CSS dan juga dibutuhkan oleh seorang Front End Developer adalah kemampuan mengelola *framework* dan *library*.

Namun semakin banyak bahasa pemrograman yang kamu kuasai, maka kesempatan untuk menjadi seorang front-end developer akan semakin terbuka lebar.



**Seorang Back End Developer** harus memiliki kemampuan dasar yaitu bisa membaca bahasa pemrograman yang ditampilkan pada "*balik layar*" sebuah situs dan aplikasi. Bahasa pemrograman yang sering digunakan di antaranya **framework** (**PHP**, GO, C#, Express dan Django/Python) dan **library**. Selain itu, MySQL, SQL Server, dan Oracle juga termasuk ke dalam skillset seorang Back End Developer.





What do you *think* about  
Framework Programming?



**There're *many* Web Development Frameworks**

**Can you build  
your own  
Programming  
Framework?**





# Many Different Languages

## Why Are There So Many Different Languages?

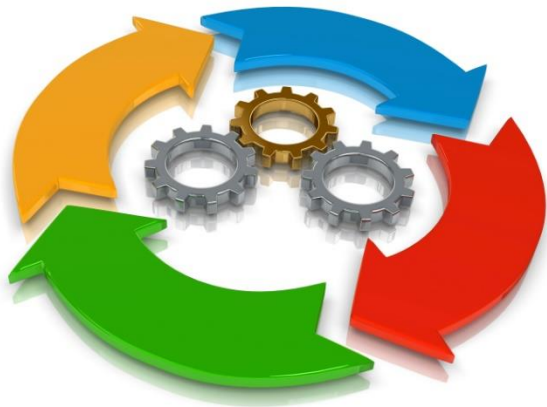


If you are just getting started with web development, you might find and realize that you might have to learn more than one programming language and it is hard enough to learn one language. *So, why are there so many different programming languages out there?*

# What Is the Purpose of a Programming Framework?



**Save Time**



**Ensure Best Practices Are Followed. Developers should learn what the state-of-the-art solution is and use that.**

# Types of Programming Frameworks

## Web Frameworks

When most people hear about framework, we might think of **web frameworks**, and there's a reason for this simple term. Such frameworks changed the way web development is designed, making it easier for millions of developers to build useful applications. So, what are web frameworks and how do they work? Web frameworks more specifically, back-end web frameworks help developers create **web applications**, also called **dynamic websites**. These are websites that you don't build with just [HTML](#), [JavaScript](#), and [CSS](#).

## Front-End Frameworks

Web frameworks can also be front-end frameworks. “*front-end*” means the **client-side part of the application**. In other words, it's what executed in the user's browser, as opposed to the back end, which is executed in a server. Front-end frameworks allow developers to leverage technologies such as [HTML](#), [CSS](#), and most importantly, [JavaScript](#) to create applications that feel as rich as desktop apps.

## Mobile Development Frameworks

Mobile application development is a gigantic market, and [\*its growth shows no signs of slowing down\*](#). So, the creation of frameworks to help in this area shouldn't come as a big surprise (**Flutter**, **Xamarin**, **Ionic** and **React Native**).

# Common Misconceptions About Frameworks

## Frameworks vs. Programming Languages

keep in mind that languages are used to write frameworks. So, David Heinemeier Hanson used the Ruby language to write the Ruby on Rails framework. Similarly, the PHP and Python languages are the foundations of the Laravel and Django frameworks, respectively.

## Frameworks vs. Libraries

What's the difference? First, understand that both terms might have different meanings in different contexts. So, the line between libraries and frameworks is often a thin one.

Keep in mind that there's an overlap between the two concepts.

A common saying people use to explain the difference between libraries and frameworks is this: ***Your code calls the library, but the framework calls your code.***

# Software Framework

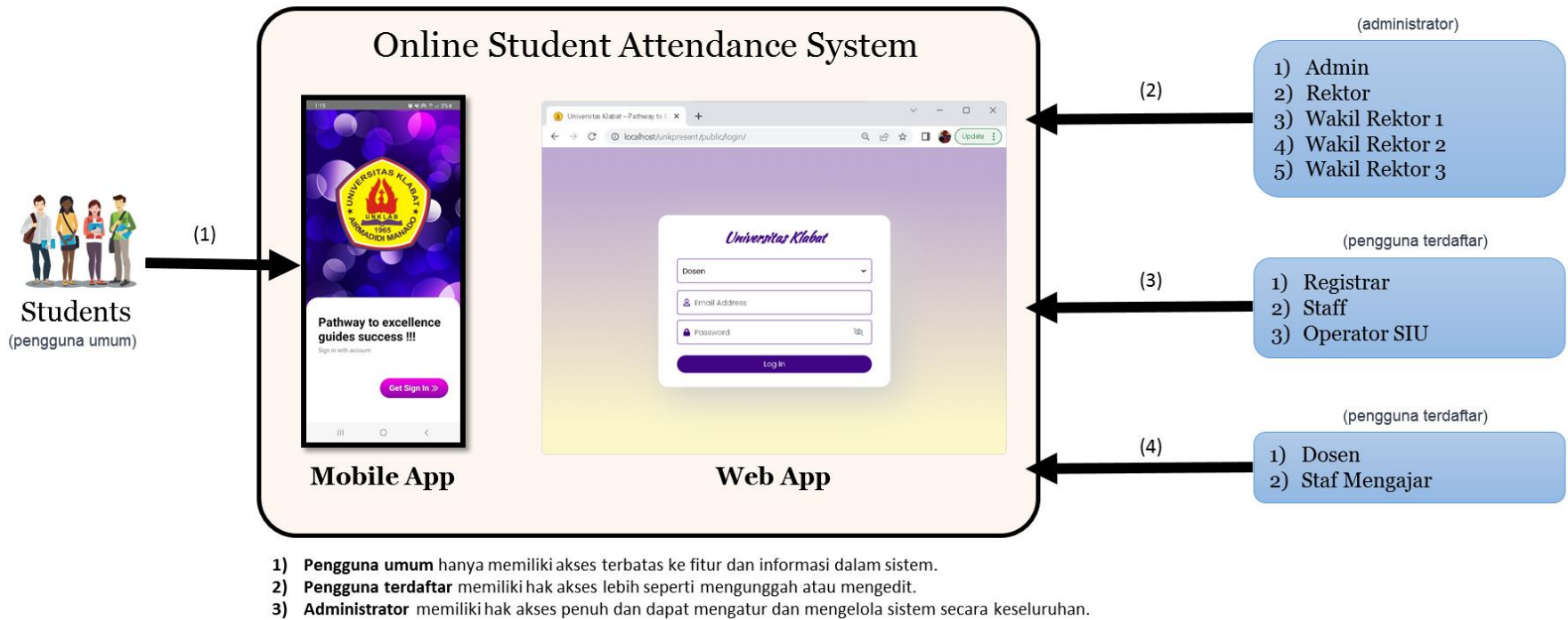
- ❑ A framework, or software framework, is a **platform** for developing *software applications*.
- ❑ It provides a foundation on which software developers can build programs for a specific platform.
- ❑ For example, a framework may include **predefined classes and functions** that **can be used to process input, manage hardware devices, and interact with system software**.
- ❑ A framework is similar to an application programming interface (API), though technically a framework includes an API. As the name suggests, a framework serves as a foundation for programming, while an API provides access to the elements supported by the framework.
- ❑ A framework may also include code **libraries**, a **compiler**, and **other programs** used in the **software development process**.

# Progressive Web Apps



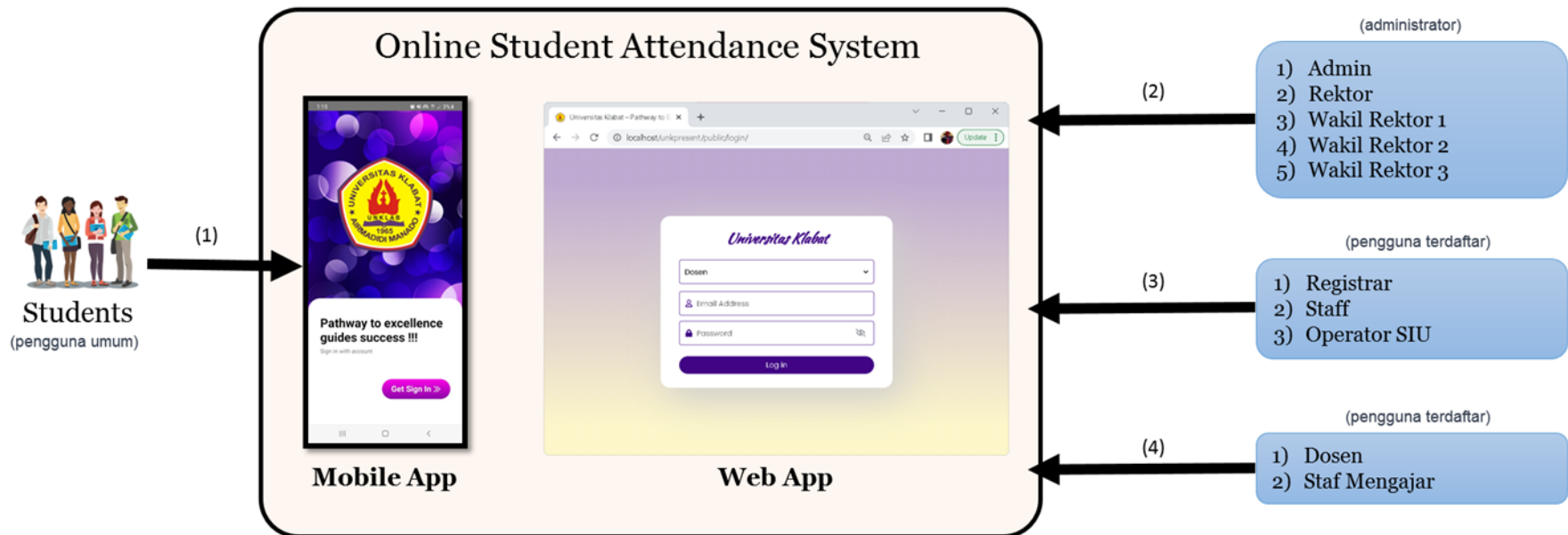
Progressive web apps enable websites to function more like native mobile apps in exchange for some flexibility. A Progressive Web App (PWA) is a web app that delivers an **app-like experience to users** by using modern web capabilities. In the end, it's just your regular website that runs in a browser with some enhancements. It gives us the ability:

- 1) To install it on a mobile home screen.
- 2) To access it when offline.
- 3) To access the camera/location.
- 4) To get push notifications.
- 5) To do background synchronization.

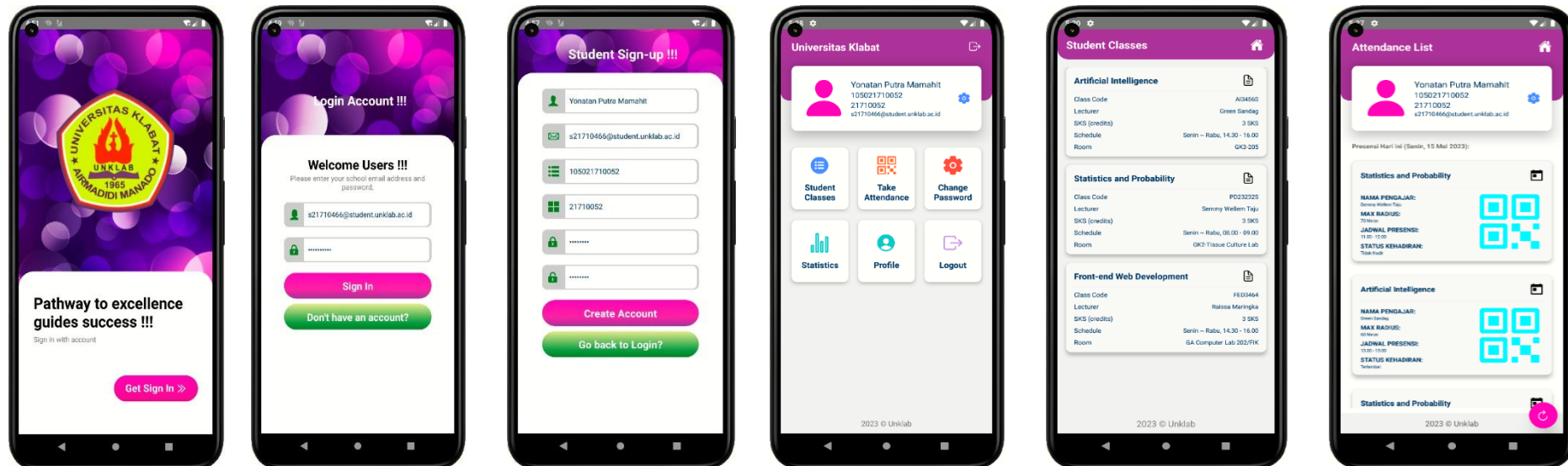


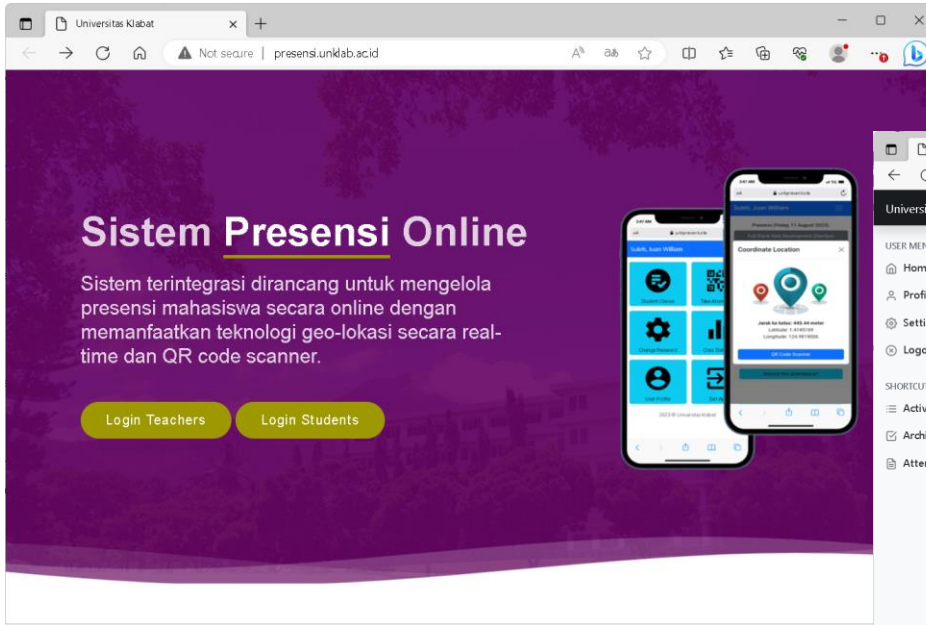
Sistem absensi mahasiswa secara online dengan dua aplikasi (*mobile app* dan *web app*) memiliki tujuan untuk memudahkan *students* dan *staf* dalam melakukan absensi dan pengelolaan kehadiran mahasiswa di kelas.



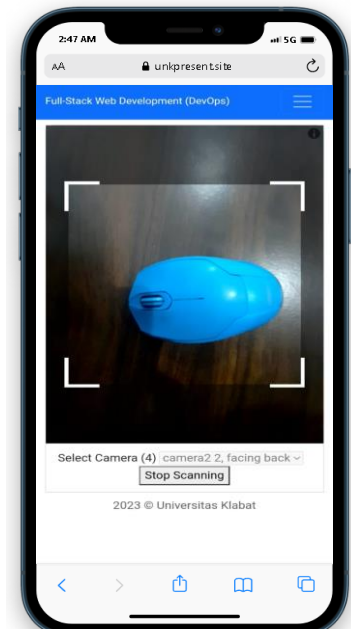
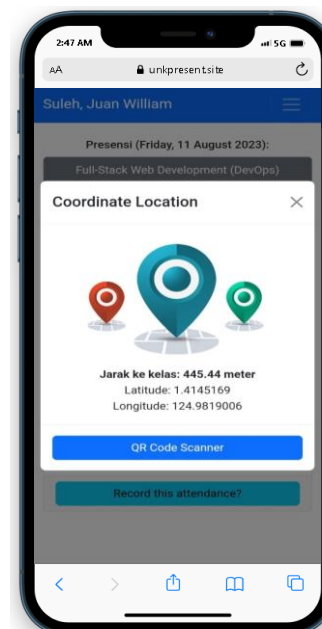
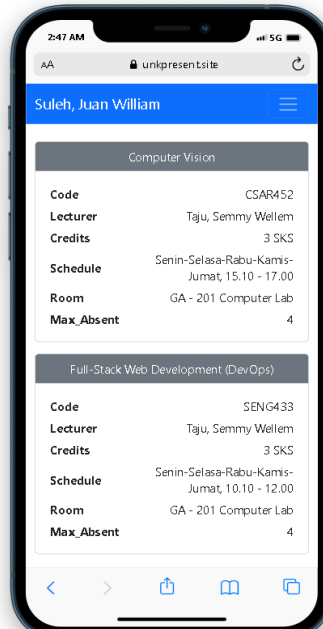
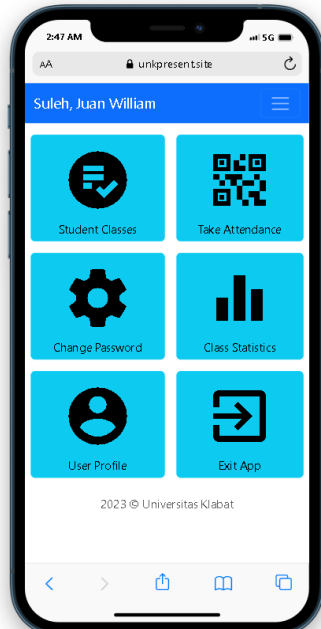
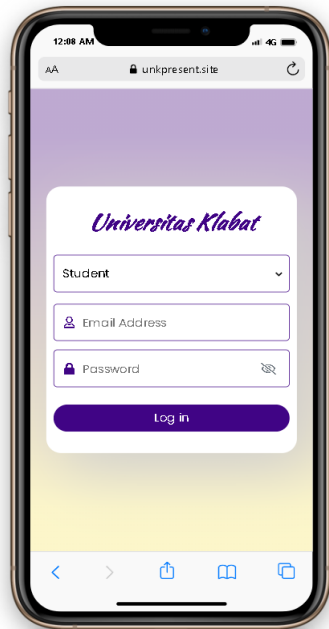


- 1) **Pengguna umum** hanya memiliki akses terbatas ke fitur dan informasi dalam sistem.
- 2) **Pengguna terdaftar** memiliki hak akses lebih seperti mengunggah atau mengedit.
- 3) **Administrator** memiliki hak akses penuh dan dapat mengatur dan mengelola sistem secara keseluruhan.





| No. | Students                                                     | Number meetings? | Hadir    | Late    | Izin    | Alpa    |
|-----|--------------------------------------------------------------|------------------|----------|---------|---------|---------|
| 1   | Abram, Mogandi Timotius<br>S2200060@student.unklab.ac.id     | 19 Meetings      | 19 times | 0 times | 0 times | 0 times |
| 2   | Amping, Bryan<br>S2200047@student.unklab.ac.id               | 19 Meetings      | 18 times | 1 times | 0 times | 0 times |
| 3   | Dumais, Ade Pricilia<br>S2200153@student.unklab.ac.id        | 19 Meetings      | 19 times | 0 times | 0 times | 0 times |
| 4   | Haerani, Christian Solafide<br>S2200045@student.unklab.ac.id | 19 Meetings      | 14 times | 3 times | 0 times | 2 times |
| 5   | Kabo, Deo Timothy<br>S2200037@student.unklab.ac.id           | 19 Meetings      | 16 times | 2 times | 1 times | 0 times |
| 6   | Korengkeng, Vito Julio<br>S2200432@student.unklab.ac.id      | 19 Meetings      | 18 times | 1 times | 0 times | 0 times |
| 7   | Mumu, Griffin Meidy<br>S2200054@student.unklab.ac.id         | 19 Meetings      | 18 times | 0 times | 0 times | 1 times |
|     | Mahut, Drivan Doter                                          |                  |          |         |         |         |



Apa tantangan seorang  
“*Back-end Developer*”  
di era digital khususnya di era  
perkembangan AI  
sekarang ini?



Seorang Back-end Developer akan menghadapi beberapa tantangan penting dalam era digital, terutama dalam konteks perkembangan AI. Meskipun Back-end Developer berfokus pada bagian yang **tidak terlihat oleh pengguna** (seperti *database*, *server*, dan *logika bisnis*), perubahan teknologi seperti AI dapat memengaruhi cara **merancang, mengembangkan** dan **mengelola** sistem.

Pada kenyataannya, profesi ini (Back-end Developer) perlu beradaptasi dengan cepat, terus meningkatkan pengetahuan, dan belajar bagaimana mengintegrasikan AI dengan cara yang efektif dalam lingkungan pengembangan *back-end* yang semakin maju.





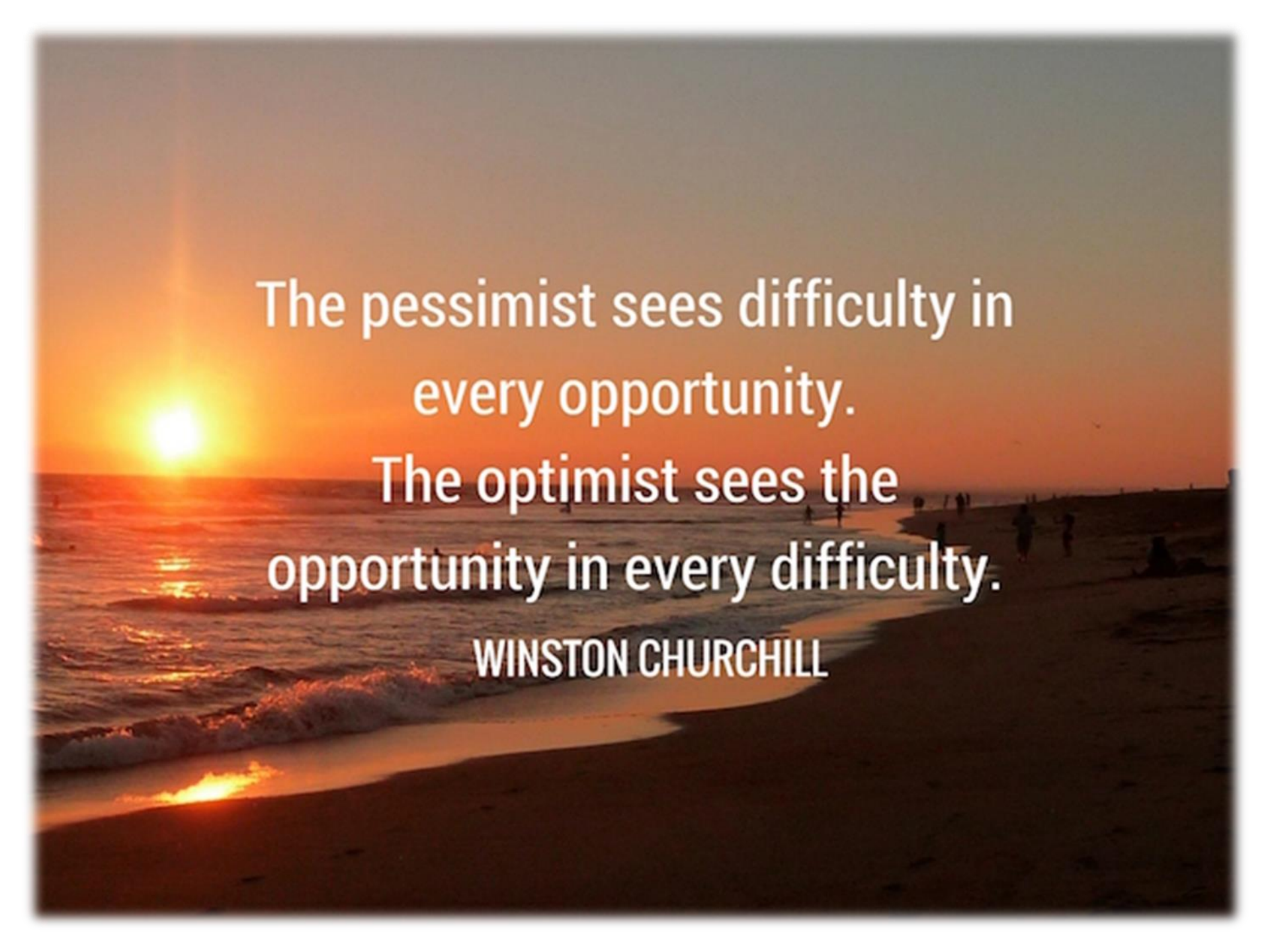
# Elon Reeve Musk FRS (*born June 28, 1971*)



- ❖ **OpenAI** is an artificial intelligence (AI) research laboratory consisting of the for-profit corporation OpenAI LP and its parent company, the non-profit OpenAI Inc. The company, considered a competitor to DeepMind, conducts research in the field of AI with the stated goal of promoting and developing friendly AI in a way that benefits humanity as a whole.
- ❖ **Neuralink** Corporation is a neurotechnology company developing implantable brain-machine interfaces (BMIs) and founded by Elon Musk and others. The company's headquarters is in San Francisco. The company was launched in 2016 and was first publicly reported in March 2017.

- ❑ Founder, CEO and chief engineer of **SpaceX**
- ❑ CEO and product architect of **Tesla, Inc.**
- ❑ Founder of The **Boring Company** and **X.com** (now part of **PayPal**)
- ❑ Co-founder of **Neuralink**, **OpenAI**, and **Zip2**
- ❑ Former chairman of **SolarCity** (now **Tesla Energy**)

"YOUR **FUTURE** IS CREATED BY WHAT  
YOU DO { **TODAY** }  
NOT ~~TOMORROW~~"

A photograph of a sunset over a beach. The sun is a bright, glowing orb on the left side of the horizon, casting a long, shimmering reflection on the water. The sky is a gradient of orange and yellow. In the foreground, the dark sand of the beach is visible, with gentle waves washing onto it. Several small, dark silhouettes of people are scattered along the shoreline in the distance.

The pessimist sees difficulty in  
every opportunity.

The optimist sees the  
opportunity in every difficulty.

WINSTON CHURCHILL





Padi yang panen hari ini tidak ditanam kemarin sore.



Di dalam kelas ini, tidak ada unsur *kompetisi*. Sebaliknya, diharapkan *kerjasama* dan saling bantu antar teman satu sama lain.

# END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

